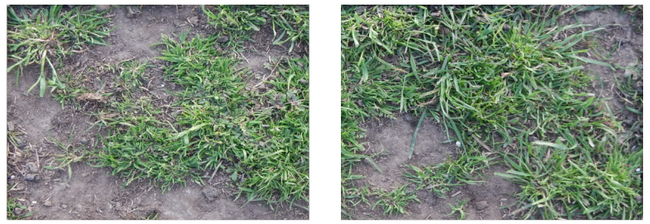
## a One pasture photograph, two complete views



CSIRO RGB photograph: 2000 × 1000



Left and right views: 1000 × 1000 each

One image-level biomass label
Views stay in their parent fold

## b Retained I-JEPA grass adaptation

30,663 GrassClover images · 2,400 updates



Visible patches

Student encoder
final 8 blocks
trainable

Predictor
trainable

Latent
prediction
loss

EMA



EMA target encoder
no optimizer gradients

Target features
stop-gradient

## c Predictor specialization, then biomass regression

CSIRO views: 224 × 224 · paired original/shuffle arms



Frozen student
encoder

Trainable
predictor

Latent
prediction loss

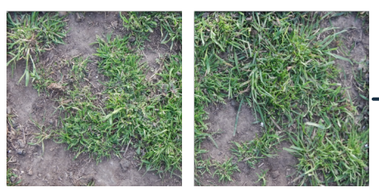
Target patches removed BEFORE student attention

Full view → frozen EMA encoder → target features

stop-gradient

After specialization: 4 target-block features → trained biomass head

## d Separate DINOv3 biomass pipeline



DINOv3 ViT-B/16
512 × 512 per view

32 × 32 patch tokens
→ 8 × 8 cells / view

Shared local
regression head

Mean of 128
cell predictions

Green
Clover
Dead

Frozen reference; final 2 / 4 / 8 blocks fine-tuned in continuation

GDM = Green + Clover; Total = GDM + Dead

CLS / register tokens excluded · 4 × 4 patch averaging

Real development photograph ID567744300 · illustrative masks, not actual sampled mask geometry · independent biomass labels remain per photograph